

Run 04 | TEA

India

IV only

Target Total early-stage entrepreneurial activity; binary 1 vs. 0

Sample 27,555 respondents **Coverage** 1 country; 2015–2022

Positive cases 3,188 (11.57%) **Case time** 33s

Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9969). Highest F1: Decision Tree (0.9347). The all-negative accuracy baseline is 88.43% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9969	0.9731	98.40	91.65	94.82	0.9321	<1s
Lasso	0.9968	0.9697	98.38	89.83	97.02	0.9328	6s
Logistic Regression	0.9968	0.9701	98.35	91.24	94.82	0.9299	<1s
Elastic Net	0.9967	0.9696	98.38	89.83	97.02	0.9328	7s
CatBoost	0.9967	0.9695	98.31	89.31	97.02	0.9300	2s
XGBoost	0.9967	0.9683	98.38	91.64	94.66	0.9313	<1s
H2O GBM	0.9966	0.9680	98.42	90.68	96.23	0.9337	2s
Ridge	0.9965	0.9674	98.38	90.18	96.55	0.9325	8s
Decision Tree	0.9941	0.9209	98.46	91.57	95.45	0.9347	<1s
Naive Bayes	0.9348	0.6794	93.39	67.84	81.48	0.7404	5s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,952.68	61.33
2	Established business owner	738.73	23.20
3	Knowing entrepreneurs	70.27	2.21
4	Business angel	63.30	1.99
5	Business exit	50.97	1.60
6	Business exit (missing indicator)	43.01	1.35
7	Self-efficacy	35.14	1.10
8	Business angel (missing indicator)	24.18	0.76

Share is the percentage of total forest impurity importance across all 25 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 04_obj1_india_iv_TEA. Scores and importance are taken directly from saved CSV outputs.